

Java Exception Handling Lab Report

Program: IllegalArgumentException

1. Aim

To demonstrate the use of IllegalArgumentException in Java for validating a deposit amount before updating an account balance.

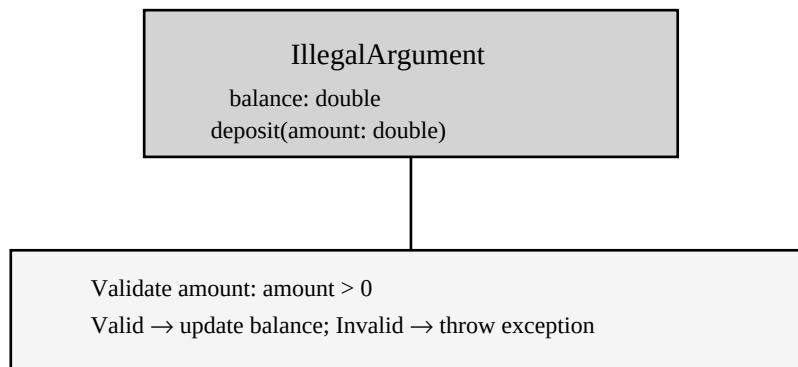
2. Description

IllegalArgumentException is an unchecked exception thrown when a method receives an argument that is invalid or inappropriate. In this program, the deposit() method checks whether the amount is less than or equal to zero. If the condition is true, an IllegalArgumentException is thrown with a descriptive message. The main method calls deposit(500), which is valid, so no exception is generated.

3. Tools Used

- Java Development Kit (JDK)
- Java compiler (javac)
- IDE or command prompt
- Operating system

4. UML / Program Structure



5. Algorithm

1. Start the program.
2. Create an account object.
3. Call the deposit() method with an amount.
4. Check whether the amount is less than or equal to zero.
5. If the amount is invalid, throw IllegalArgumentException with an error message.
6. If the amount is valid, add it to the account balance.
7. Handle the exception in the catch block if it occurs.
8. Stop the program.

6. Complete Java Source Code

```
package exceptions;
public class IllegalArgument{
    private double balance;

    public void deposit(double amount) {
        // Validate input state
        if (amount <= 0) {
            throw new IllegalArgumentException("Deposit amount must be greater than zero. Provided: " + amo
        }
        this.balance += amount;
    }

    public static void main(String[] args) {
        IllegalArgument account = new IllegalArgument();
        try {
            account.deposit(500);
        } catch (IllegalArgumentException e) {
            System.out.println("Invalid operation: " + e.getMessage());
        }
    }
}
```

7. Sample Output

No output is displayed because the deposit amount 500 is valid and no exception is thrown.

(No console output)

8. Result

The program successfully demonstrated input validation using `IllegalArgumentException`. Since the deposit amount is 500, the balance is updated and the catch block is not executed.